High Value Customers and Top-Selling Products

Business Question: Which cookie products and customer segments generate the highest revenue and repeat purchases, and how can the company optimize sales strategy based on this insight?

Why This Matters? Helps identify which customers and products to focus on, therefore maximizing sales year-round.

Data Description:

Source: How to use Power BI DAX – Tutorial by Kevin Stratvert on YouTube – Excel files:
https://onedrive.live.com/?redeem=aHR0cHM6Ly8xZHJ2Lm1zL3UvcyFBbXhyb2ZaWmxaLXdoTUlnUzlXSlM3QndTZkhCYnc%5FZT0wejNuUW4&id=B09F9559F6A16B6C%2174016&cid=B09F9559F6A16B6C

- Table 1 – Cookie Types:
  - Number of Records: 7
  - Fields (used fields):
    - Cookie type
    - Units sold
    - Revenue per cookie
    - Cost per cookie

- Table 2 – Cookie Types:
  - Number of Records: 5
  - Fields (used fields):
    - Customer ID
    - Name

- Table 3 – Cookie Types:
  - Number of Records: 700
  - Fields (used fields):
    - Customer ID
    - Order ID
    - Product
    - Units sold
    - Date
    - Revenue
    - Cost

Tools & Technologies Used:

- Microsoft Excel: data cleaning, verification and empty cells checking

- MySQL Workbench: General insights throughout the years, querying, EDA
- Microsoft Power BI: identify trends, make charts, DAX calculations

Exploratory Data Analysis (EDA):

- Trends from Sep 2019 - Dec 2020: October generated the most orders
- best-selling product (order count & revenue) is chocolate chip cookies, least ordered is fortune cookies
- ACME Bites generated the highest revenue, Tres Delicious generated the least

Techniques used: SQL querying and Power BI visualizations

Detailed SQL Queries:

- What is the total revenue per customer?

Code: SELECT c.CustomerID, Name, sum(Revenue) as Revenue

FROM customers as c

JOIN orders as o ON c.CustomerID = o.CustomerID

group by c.CustomerID, Name

order by sum(Revenue) desc

- What is the total revenue per product?

Code: SELECT Product, SUM(UnitsSold) AS "Units Sold", SUM(Revenue) AS "Revenue"
FROM orders
GROUP BY Product
ORDER BY SUM(Revenue) DESC

- what is the monthly orders per product?

Code: SELECT Product, MONTH(Date) as Month, (COUNT(OrderID)) AS Orders
FROM orders
GROUP BY Product, Date
ORDER BY MONTH(Date) ASC

Used: SELECT, FROM, JOIN, GROUP BY, ORDER BY, SUM, COUNT, string function time MONTH.

Data Visualization and Report Creation:

- Tables of name, order count, units sold and revenue of both cookie products and customers for 2020.
- Line and clustered column chart showing the monthly revenue and count of orders for oatmeal raisin cookies for ACME Bites in 2020 (plus chocolate chip & cookies in a slicer)

- Line and clustered column chart showing the monthly revenue and count of orders for white chocolate macadamia nut cookies for Wholesome foods in 2020 (plus chocolate chip cookies in a slicer)

DAX measurements:

Calculated columns:

Differentiate between orders with chocolate chip cookies:

Choc chip or no = if('Orders'[Product] = "Chocolate Chip", "Chocolate Chip", "Not Chocolate Chip")

What is the percent difference in revenue of chocolate chip cookies vs other products in 2020?

- **Avg rev choc chip =** CALCULATE(AVERAGE(Orders[Revenue]), Orders[Choc chip or no] = "Chocolate Chip")
- **Avg rev non-choc chip =** CALCULATE(AVERAGE(Orders[Revenue]), Orders[Choc chip or no] = "Not Chocolate Chip")

Code: % diff rev CC to NCC = ([Avg rev choc chip] - [Avg rev non-choc chip]) / [Avg rev non-choc chip]

Used: IF Statement, CALCULATE, AVERAGE

Key Insights:

- ACME Bites generates the **83.71%** more revenue and makes **69.11%** more orders, followed by Wholesome Foods.
- Chocolate Chip is the leading product generating **37.41%** higher revenue.
- Oatmeal Raisin is second best for ACME Bites.
- White Chocolate Macadamia Nut was second for Wholesome Foods, **26.7%** higher revenue.
- ACME Bites sales for both products peak mid-year, with Chocolate Chip having **52%** higher monthly orders.
- Chocolate Chip peaks in December for Wholesome Foods, while White Chocolate Macadamia Nut peaks mid-year but is less frequently ordered.

Recommendations:

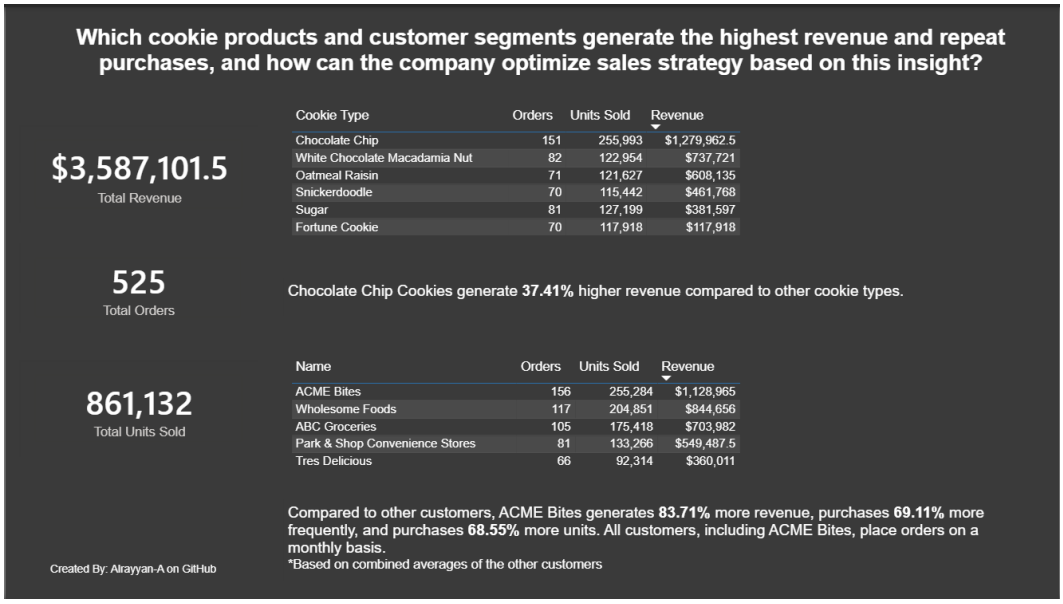
- **Promotions** for Chocolate Chip Cookies in **Q1 & Q4** should be done for ACME Bites since most sales occur mid-year.
- For year-round orders of White chocolate Chip Cookies for Wholesome Foods, **promotions in Jan, Mar, May, Aug & Sep** should be done.

Conclusion:

A cookie-company dataset with multiple tables on cookie types, customers and orders by Kevin Stratvert was analyzed using MS Excel, MySQL and MS Power BI to determine the customers and products that generate the highest revenue and orders.

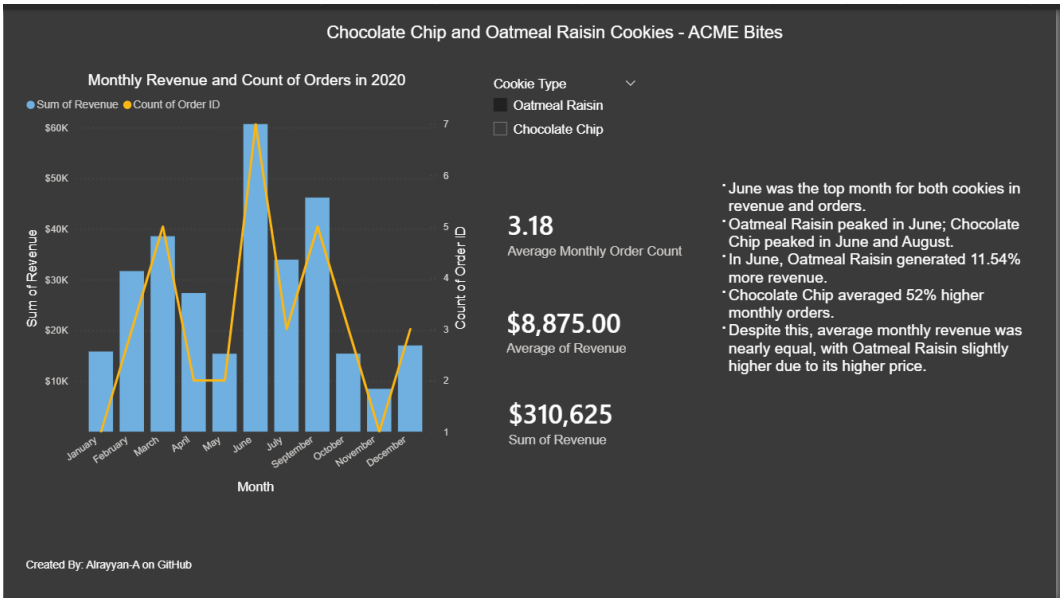
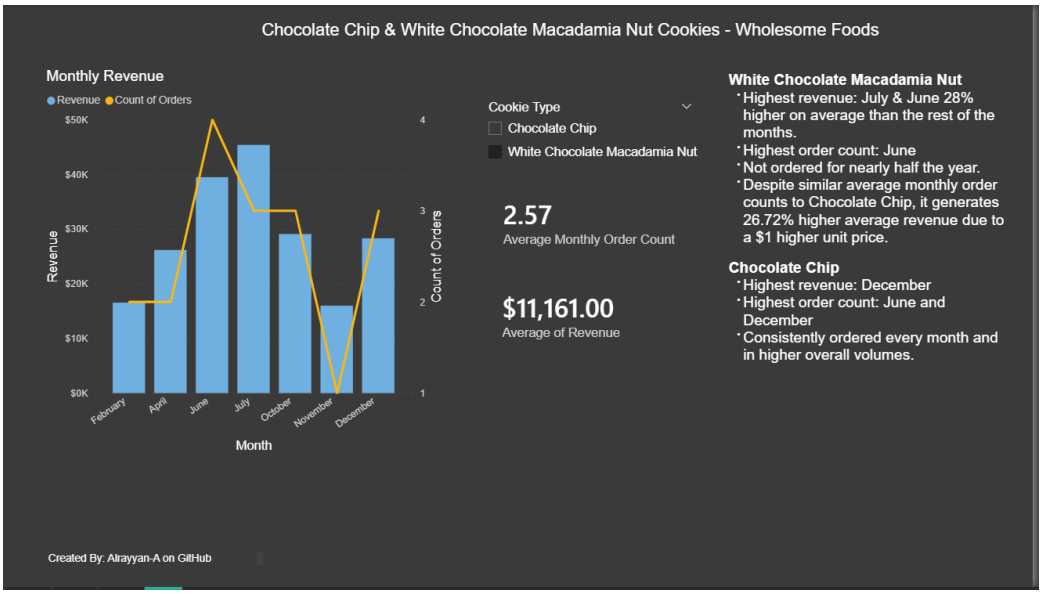
After the analysis, no.1 customer is ACME Bites with chocolate chip and oatmeal raisin as best sellers especially mid-year. No. 2 customer is Wholesome foods with chocolate chip and white chocolate macadamia nut cookies, where sales peak in December and mid-year, respectively. To drive sales year-round, create targeted promotions for the top customers in the times of year where the sales for the top product are low. This is important because helps create effective sales strategies that makes sales happen more consistently.

Screenshots of Report:



Cookie Type	Orders	Units Sold	Revenue
Chocolate Chip	151	255,993	$1,279,962.5
White Chocolate Macadamia Nut	82	122,954	$737,721
Oatmeal Raisin	71	121,627	$608,135
Snickerdoodle	70	115,442	$461,768
Sugar	81	127,199	$381,597
Fortune Cookie	70	117,918	$117,918

Name	Orders	Units Sold	Revenue
ACME Bites	156	255,284	$1,128,965
Wholesome Foods	117	204,851	$844,656
ABC Groceries	105	175,418	$703,982
Park & Shop Convenience Stores	81	133,266	$549,487.5
Tres Delicious	66	92,314	$360,011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